Stream Count Predictive Analysis for Upcoming Songs on Spotify using Machine Learning: A Systematic Literature Review

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Stream Count Predictive Analysis for Upcoming Songs on Spotify using Machine Learning: A Systematic Literature Review

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Background of Research

01 BACKGROUND OF RESEARCH

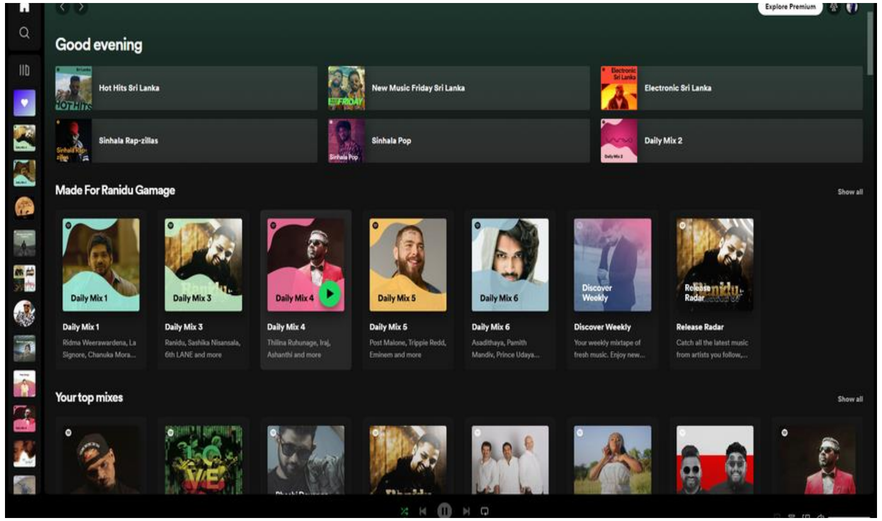
- Music streaming industry (Spotify)

- streaming platforms that provide global reach to artists and connect directly to millions of listeners.
- Streaming platforms include Apple Music, SoundCloud, Amazon Music Unlimited, and YouTube Music.
- Artists can make money in the music industry in many ways.
- there are 551 million listeners on Spotify.

BACKGROUND OF RESEARCH CONT'D

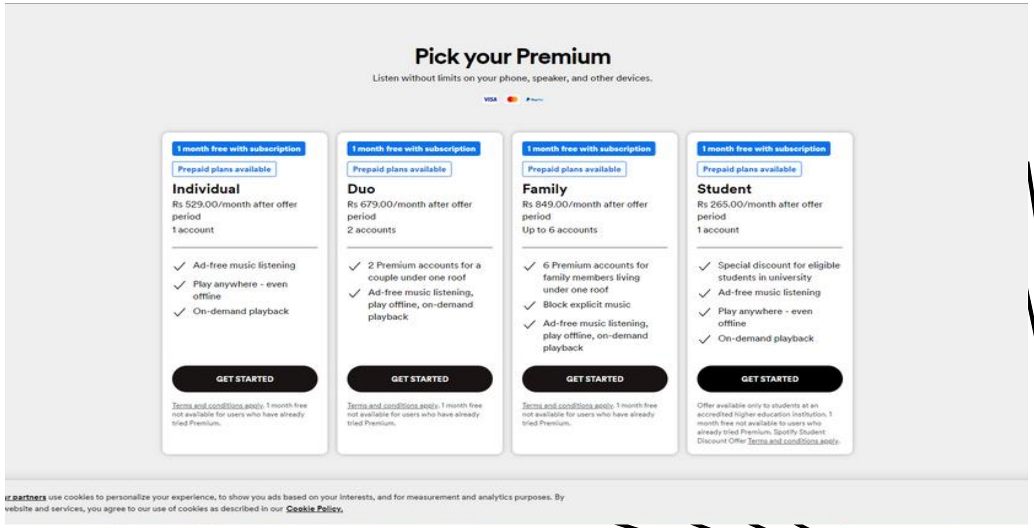
Playlist sites

selected popular playlists allow their music to be made available to a wider audience and can lead to increased traffic.



Streaming Royalties

Each time a user listens to a song, the artist receives a small portion of the payment.



BACKGROUND OF RESEARCH CONT'D

- Factors influencing stream count

many factors that affect the number of times a song is streamed on a music streaming platform. For artists, labels, and marketing professionals, understanding these factors is critical to achieving traffic numbers and overall success.

- Popularity
- Playlist layout
- Airtime and marketing
- Genres
- Collaborations and features
- Virality and cultural impact
- User-generated content and social media
- Spotify audio track features

LITERATURE REVIEW

Literature Review

Identification

- IEEE Xplore
- Research Gate
- EBSCO
- Google Scholar
- Scopus

↓

Screening

- Duplications excluded
- Articles excluded after the initial title, abstract screening, and conclusions

↓

Eligibility

- Full-text articles were assessed
- Irrelevant, duplicated, no full-text available articles were excluded

↓

Inclusion

- Analyze the Papers
  - Predictive Analytics
  - Music Streaming
  - Spotify, Stream Count Prediction

RESEARCH 01 - PREDICTING MUSIC POPULARITY USING MACHINE LEARNING ALGORITHM AND MUSIC METRICS AVAILABLE IN SPOTIFY

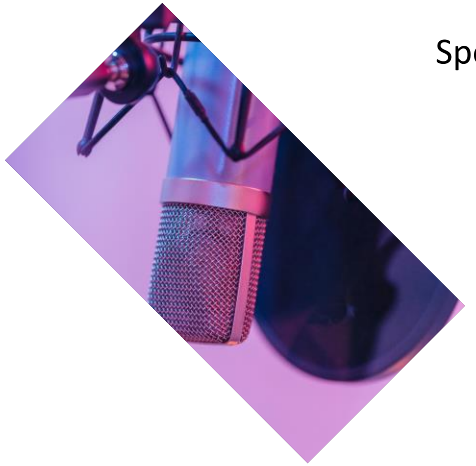
- The paper focuses on predicting the popularity of songs using machine learning algorithms and music metrics available on Spotify.

Data set

- collected from Kaggle and extracted from the Spotify API.

Methods used

- Random Forest classifier
- K-Nearest Neighbor classifier
- Linear Support Vector classifier



Results and Findings

- The Random Forest classifier is found to be the most effective in predicting song popularity, with high accuracy, precision, recall, and F1-score.

RESEARCH 02 - Predicting a Hit Song with Machine Learning: Is there an apriori secret formula?

- The paper aims to predict the success of songs by analyzing their technical parameters and sentimental analysis of the lyrics using machine learning algorithms.

Data set

- The dataset consists of 310 hit songs and 337 non-hit songs.

Methods used

- Logistic Regression
- Decision Trees
- Naïve Bayes
- Random Forests

Results and Findings

- Danceability was identified as the most critical feature in determining the success of a song,

RESEARCH 03 - Performance Prediction of Songs on Online Music Platforms

- The paper focuses on predicting the performance of songs on the Billboard Hot 100 charts using objective and well-defined features obtained from Spotify's Web API.

Data set

- The researchers collected data from Spotify's Web API to obtain objective and well-defined features of songs.

Results and Findings

- The Random Forest Classifier showed the best accuracy of 56.2% for predicting the Top 10-40 classification, while the XGBoost Classifier achieved an accuracy of 82.0% for predicting the Top 10 classification.

Methods used

- Regression models: Ridge Regression, Random Forest Regression, and XGBoost Regression
- Classification models: Single-class and multi-class classifiers were used to predict a song's rank bucket.
- Ensemble models
- Cross-validation
- Feature encoding
- Evaluation metrics

RESEARCH 04 - HIT SONG PREDICTION: LEVERAGING LOW- AND HIGH-LEVEL AUDIO FEATURES

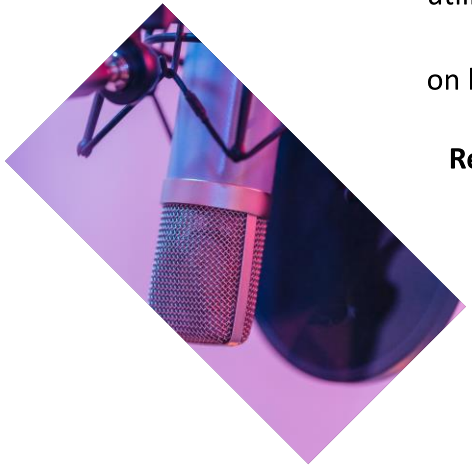
- The paper aims to predict the potential success of a song based on its acoustic characteristics using a wide and deep neural network architecture that combines low-and high-level audio features.

Data set

- utilizes the Million Song Dataset (MSD) for its experiments on hit song prediction.

Methods used

- The paper utilizes a wide and deep neural network architecture for hit song prediction.



Results and Findings

- Experiment 1: wide and deep network architecture achieved better performance than a linear regression baseline and utilized the two parts of the network individually.
- Experiment 2: relative importance of individual feature subsets in the wide and deep neural network. Different feature sets (low- and high-level) were experimented with, and their prediction performance was compared

RESEARCH 05 - Evolution of global music trends: An exploratory and predictive approach based on Spotify data

- The paper analyzes the most popular songs from 52 countries on Spotify, focusing on their features of danceability, positivity, and intensity.

Data set

- the most popular songs from 52 countries on Spotify focusing on their features

Results and Findings

- most popular songs from 52 countries on Spotify revealed behavioral changes in music listening patterns due to the COVID-19 pandemic.
- The multivariate time series model proposed in the paper enabled the prediction of the type of music listened to in three different groups of countries with an error rate of around 1%.

Methods used

- multivariate time series model to predict the preferred type of music in different countries
- Time series clustering is used as an unsupervised data mining technique to organize data points into groups based on their similarity.
- Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Average Prediction Error (MAPE) to evaluate the prediction models

RESEARCH 06 - Effect of Feature Selection on the Accuracy of Music Popularity Classification Using Machine Learning Algorithms

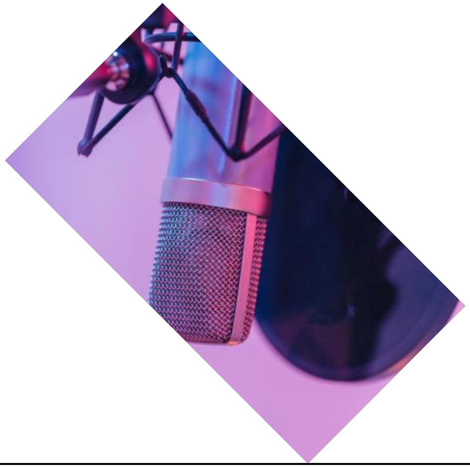
- The research analyzes the impact of feature selection on the accuracy of music popularity classification using machine learning algorithms.

Data set

- the Spotify Music Dataset, which was downloaded from the website www.kaggle.com.

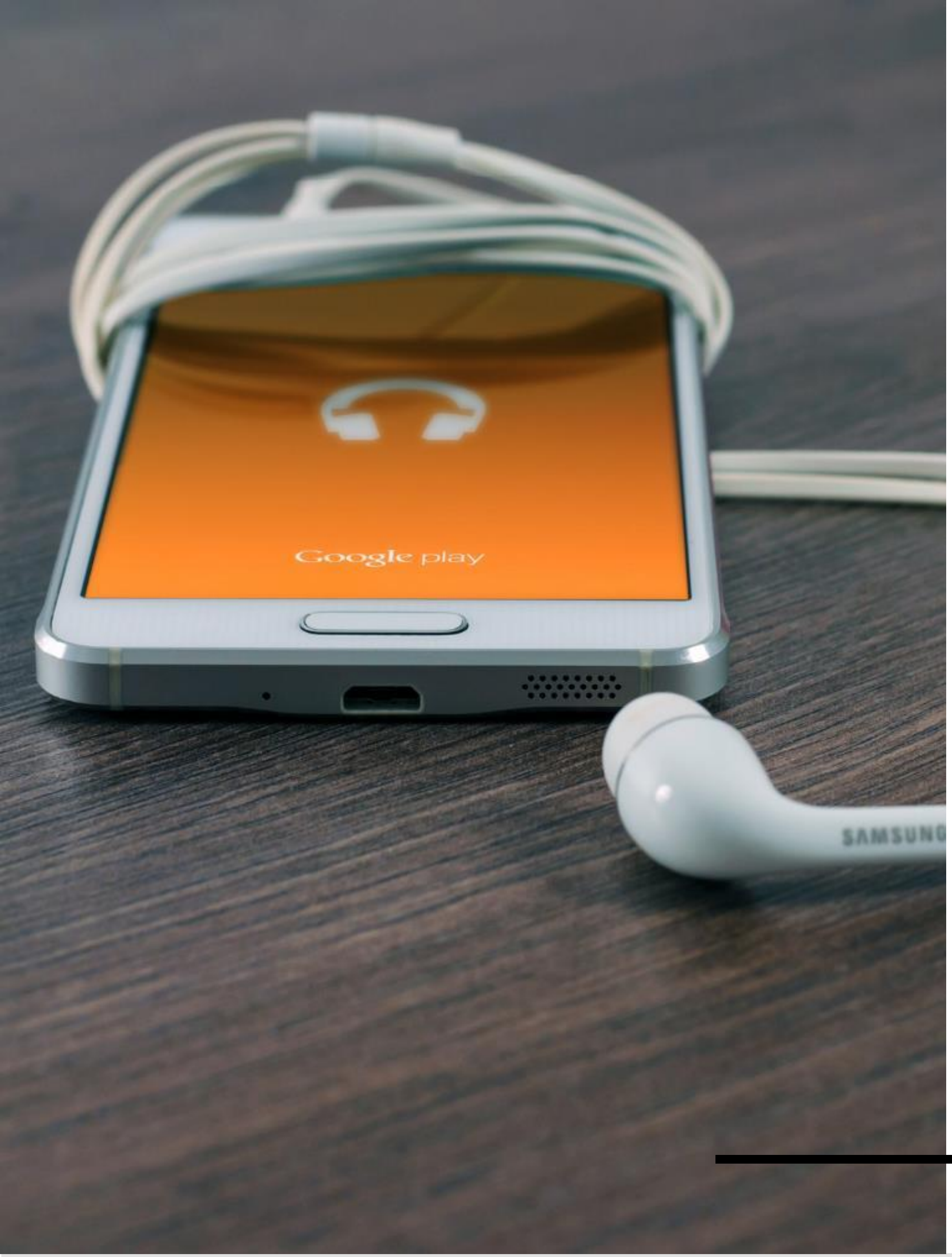
Methods used

- classification and feature selection
- logistic regression, K-nearest neighbors (KNN), and decision trees (random forest)



Results and Findings

- The paper evaluates the effect of feature selection on the accuracy of music popularity classification using machine learning algorithms.
- Machine learning algorithms using all features achieved an accuracy of 95.15%, while algorithms using features selected by feature selection achieved 95.14% accuracy.

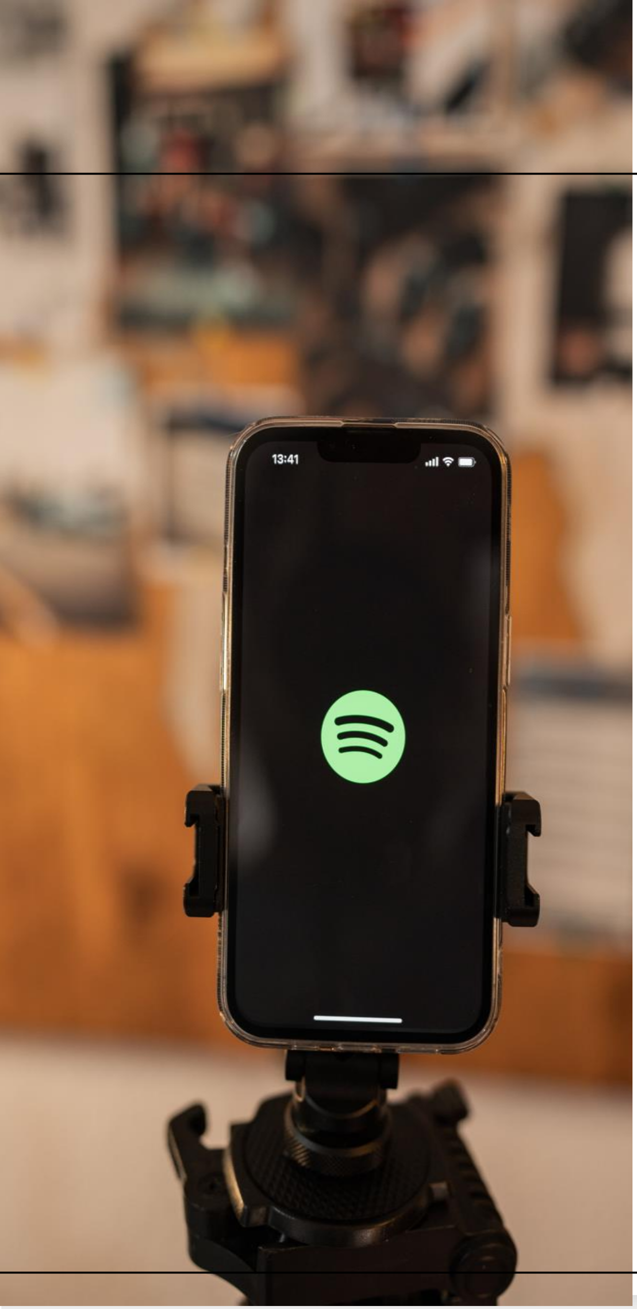


Methodology

METHODOLOGY

Methodical Review of Literature

- Identify relevant literature sources on factors influencing stream counts in the music-streaming industry.
- Conduct a comprehensive review of existing studies, articles, and reports.
- Analyze the findings to identify common themes and insights.



METHODOLOGY

Identifying the Factors

Artist popularity

size of an artist's fan base play a major role in stream count

Spotify Developer API the endpoint:

https://api.spotify.com/v1/artists/{id}

Genres

Different types have different audiences, and understanding your audience is key to optimizing

Spotify Developer API the endpoint:

https://api.spotify.com/v1/tracks/{id}

Featured artist popularity

Collaborations with other artists or featuring popular artists in a song can lead to streams

Spotify Developer API the endpoint:

https://api.spotify.com/v1/artists/{id}

User-generated content and social media

Sharing and creating user-generated content such as covers, remixes, and fan videos can generate a lot of traffic.

- TikTok

Spotify audio track features

specific attributes help Spotify understand the musical characteristics: attributes acousticness, danceability, energy, durations, instrumentalness, key, liveness, loudness, mode, speechiness, tempo, time signature, valence

Spotify Developer API the endpoint: https://api.spotify.com/v1/audio-features

METHODOLOGY

Initial Model Development and Validation

- Use the collected data to develop an initial machine-learning model.
- Evaluate the model's performance by comparing predicted stream counts with actual stream counts.

to predict the stream count

Regression Techniques Used

- Ridge Regression
- Linear Regression
- Random Forest
- Support Vector Regression
- Gradient Boosting

Regression model validation

- SMAPE

predict whether the song will be a hit or not

Classification techniques used

- Logistic Regression
- Random Forest
- Support Vector Machine
- K-Nearest Neighbor
- Gradient Boosting

Classification model validation

- Confusion Matrix - Evaluation

Results and Future Works

Streaming offers mobility and lower costs.

Results and Future Works

- Identified research gap: Lack of tailored predictive models for Sri Lanka's diverse music preferences.
- Existing studies primarily focus on global or platform-specific data, ignoring region-specific analyses.
- Challenges in incorporating local features and cultural impact for accurate stream count predictions.

- Develop tailored predictive models for Sri Lanka's music landscape by incorporating local features and cultural impact.
- Explore region-specific analyses to understand the unique musical preferences and trends of Sri Lankan audiences.
- Investigate the influence of local factors such as cultural impact, languages, and emerging artists on stream count predictions.
- Analyze the algorithms and factors influencing song rankings on popular streaming platforms used in Sri Lanka

Conclusion

Conclusion

- Predictive analytics for stream counts on platforms like Spotify play a vital role in shaping music industry strategies.
- There's a notable gap in tailored predictive models for Sri Lanka's diverse music landscape.
- Region-specific analyses and the inclusion of cultural impact are crucial for ensuring accurate predictions.
- Existing research highlights the potential for developing refined predictive models by integrating local features and platform-specific strategies.
- Future research should focus on integrating local elements and regional trends to empower the Sri Lankan music industry with more nuanced predictive models.

Q &A

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THANKS